

$$CF_3 = 142.64$$

CF_0 refers to the initial investment of USD100 million made at the beginning of the first four-month interval on 1 January. CF_1 refers to the cash flows made at end of the first four-month interval or the beginning of the second four-month interval on 1 May. Those cash flows include a cash inflow of USD2 million for the dividend received and cash outflows of USD22 million for the dividend reinvested and additional investment, respectively. The second four-month interval had no cash flow so CF_2 is equal to zero. CF_3 refers to the cash inflows at the end of the third four-month interval. Those cash inflows include a USD2.64 million dividend received and the fund's terminal market value of USD140 million.

Using a spreadsheet or IRR-enabled calculator, we use -100, -20, 0, and USD142.64 for the $t = 0$, $t = 1$, $t = 2$, and $t = 3$ net cash flows, respectively. Using either tool, we get a four-month IRR of 6.28 percent.

$$\frac{CF_0}{(1 + IRR)^0} + \frac{CF_1}{(1 + IRR)^1} + \frac{CF_2}{(1 + IRR)^2} + \frac{CF_3}{(1 + IRR)^3} = 0$$

$$\frac{-100}{1} + \frac{-20}{(1 + IRR)^1} + \frac{0}{(1 + IRR)^2} + \frac{142.64}{(1 + IRR)^3} = 0$$

$$IRR = 6.28\%$$

The quick way to annualize this four-month return is to multiply it by 3. A more accurate way is to compute it on a compounded basis as: $(1.0628)^3 - 1 = 0.2005$ or 20.05 percent.

These calculations can also be performed using Excel using the =IRR(cash flows) function, which results in an IRR of 6.28 percent.

- Interpret the differences between the Fund's time-weighted and money-weighted rates of return.

Solution:

In this example, the time-weighted return (21.03 percent) is greater than the money-weighted return (20.05 percent). The Walbright Fund's performance was relatively poorer during the eight-month period, when the fund had more money invested, than the overall performance. This fact is reflected in a lower money-weighted rate of return compared with the time-weighted rate of return, as the money-weighted return is sensitive to the timing and amount of withdrawals and additions to the portfolio.

The accurate measurement of portfolio returns is important to the process of evaluating portfolio managers. In addition to considering returns, however, analysts must also weigh risk. When we worked through Example 9, we stopped short of suggesting that in-house management was superior to Super Trust because it earned a higher time-weighted rate of return. A judgment as to whether performance was "better" or "worse" must include the risk dimension, which will be covered later in your study materials.

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ANNUALIZED RETURN



calculate and interpret annualized return measures and continuously compounded returns, and describe their appropriate uses